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Title: Leveraging Artificial Intelligence for Enhanced Fraud Detection in Financial Auditing

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Summary

This paper explores how artificial intelligence (AI) can significantly improve fraud detection in financial auditing by enhancing the ability to identify irregularities in large and complex datasets. The authors, through a combination of machine learning model evaluations and real-world case studies, demonstrate that AI-powered tools outperform traditional auditing methods in detecting financial fraud, such as revenue manipulation and expense misreporting. The study emphasizes AI's capacity to analyze patterns, detect anomalies, and predict potential fraud risks in real time, offering a scalable solution for modern audit challenges.

The methodology involves testing three AI models, decision trees, neural networks, and anomaly detection algorithms on financial datasets from 15 publicly listed companies (2018-2022), alongside qualitative insights from audit practitioners. The results show that AI models achieve a fraud detection accuracy of 85-92%, compared to 70-75% for manual audits, with neural networks excelling in identifying complex fraud patterns. A case study of a retail firm illustrates how AI flagged suspicious transactions missed by traditional audits, preventing a potential \$2 million loss. Challenges include the need for high-quality data, computational costs, and ensuring explainability of AI outputs for regulatory compliance.

The paper concludes that AI-driven fraud detection not only boosts accuracy but also reduces audit time by up to 25%, allowing auditors to focus on strategic oversight. However, it highlights barriers like integration with legacy systems and ethical concerns about algorithmic transparency. The authors advocate for hybrid AI-human auditing frameworks and standardized protocols to address these issues, positioning AI as a critical tool for enhancing trust and reliability in financial reporting.